Probing the Lower Limits of Detectable Central Black Hole Masses in Virgo Cluster CSS with JWST NIRSpec IFU Kinematics

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Abstract

The central black hole (BH) mass function remains poorly constrained as a result of observational barriers in the context of dynamical BH mass recovery at the low mass end. Constraints on BH masses in compact stellar systems (CSS) are crucial for understanding both the local universe BH mass function and formation channels for the seeds of supermassive black holes (SMBHs). The detection of central BHs in CSS, such as ultracompact dwarf galaxies (UCDs), nuclear star clusters (NSCs), and compact ellipticals (cEs), can inform and validate theories regarding the formation of BH seeds. We have obtained JWST/NIRSpec IFU kinematics of 18 Virgo cluster CSSs, the first wide-scale survey of CSS BHs. In preparation for modeling the kinematic data we have constructed mock kinematics for UCDs and CEs and used the Schwarzschild modeling code FORSTAND to find the minimum SMBH mass that can be reliably constrained at the distance of the Virgo cluster systems that have been observed. Our mock CSS span a stellar mass range of 10^7-$10^{9.5}$ M_{sun} and a broader range of stellar effective radii, designed to mimic the general population of CSS. Kinematical data for our mock CSS population was also constructed to spatially and kinematically mimic JWST/NIRspec IFU data. Modeling has been performed for isotropic and radially anisotropic velocity distributions and noise for each mock CSS. We find that the general mass limit for BH detection in Virgo cluster CSS is $M_{BH} > \sim 1\%\ M_{BH}/M_*$.

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Mock Sample

$$\rho = \rho_0(\tilde{r}/a)^{\gamma}\left[1+(\tilde{r}/a)^{\alpha}\right]^{(\gamma-\beta)/\alpha}\exp\left[-(\tilde{r}/r_{\rm cut})^{\zeta}\right]$$

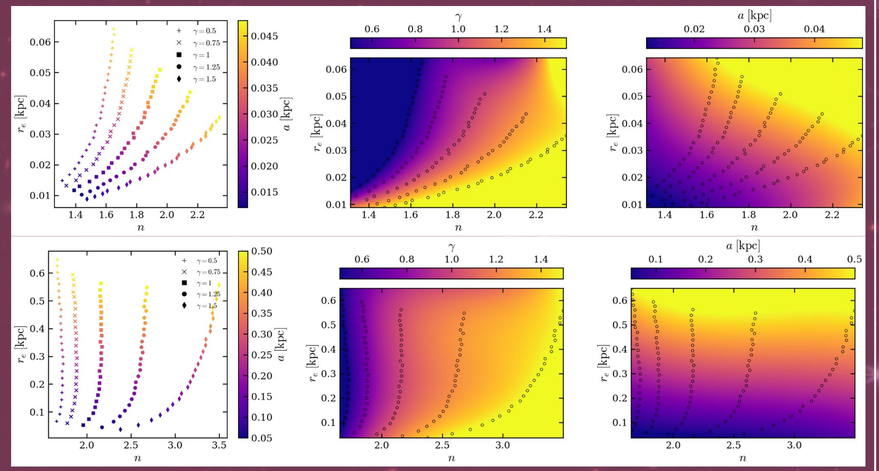

Figure 1: Mock sample for UCDs (top) and cEs (bottom). **Left:** Effective radius as a function of Sersic index, colored by scale radius *a*. **Middle & Right:** The same, but overlaid over a 2D plane constructed with third-order B-spline interpolation. This allows one to select an accurate γ and *a* based on observable parameters.

3D density model

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Two free parameters (γ and *a*)

↓

Make many mocks over range of suitable γ and *a*, fix everything else

↓

Project each *N*-body snapshot, fit 2D Sérsic to find *n* an r_e

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Use observable quantities (*n* and r_e) to obtain non-observable quantities (γ and *a*) for each target

↓

Construct accurate mocks based on determined γ and *a*

Example Models

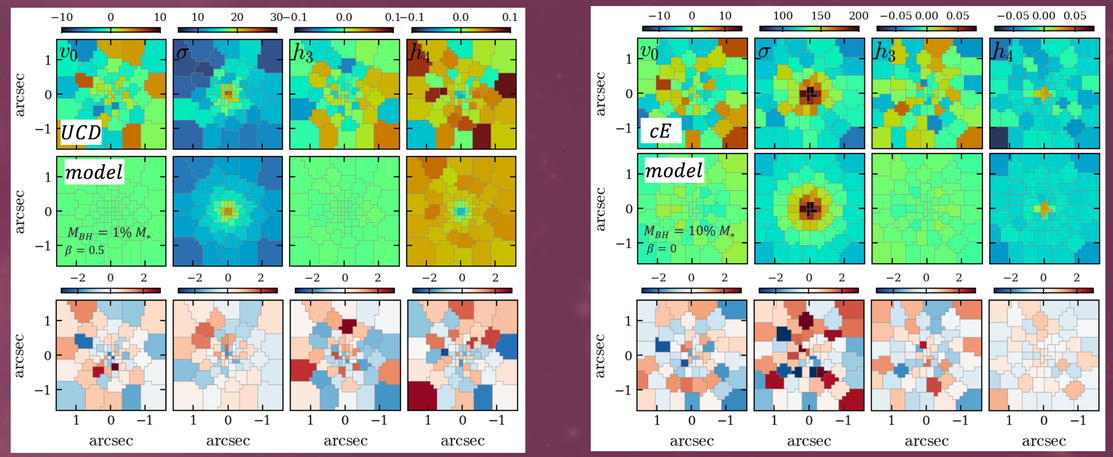

Figure 2: Mock observations of a UCD (left) and cE (right). Top row shows mock observations (line of sight velocity V_{los}, velocity dispersion σ, and third/fourth Gauss-Hermite moments h_3 and h_4). Middle row shows best fit kinematics from FORSTAND code using mock observations as constraints. Bottom panel is χ^2 error of model.

Identifying a Lower Limit of Recoverable Mass

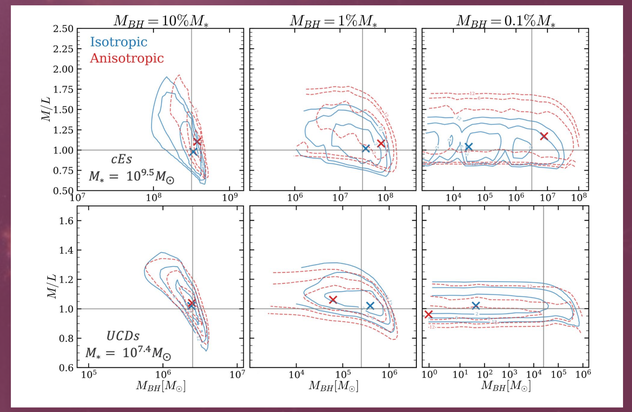

Figure 3: **Top:** The χ^2 plane for cEs with $M_{BH}/M_* =$ 0.10, 0.01, and 0.001 respectively. True M/L and M_{BH} values for each mock are denoted with the dashed black cross. For systems with a large M_{BH}/M_*, the difference in best-fit values between anisotropic and isotropic realizations is minimal, but the disparity becomes large for systems with $M_{BH}/M_* = 0.01\%$ where only an upper limit can be obtained. **Bottom:** The same, but for UCDs. The same behavior between anisotropic and isotropic realizations can be seen, however the realization with the lowest M_{BH}/M_* at the very least provides a reasonable upper limit to M_{BH}.

References *Vasiliev, E., & Valluri, M. 2020, ApJ, 889, 39; Schwarzschild, M. 1979, ApJ, 232, 236, Vasiliev E., 2019, MNRAS, 482, 1525; Liu C., et al., 2020, ApJ, 250, 17*